

$^{34}\text{S}(\text{d},\text{p}\gamma)$ 1972Fr11,1975VaYG,1970Bu18

1972Fr11: 3.50- and 4.00-MeV deuteron beams were produced from the 4-MV Van de Graaff accelerator at the Nuclear Research Center of Strasbourg, France. Targets were antimony sulphide enriched to 85.6%, $550\text{ }\mu\text{g}/\text{cm}^2$ on a 0.03 mm molybdenum layer and $150\text{ }\mu\text{g}/\text{cm}^2$ on a 0.05 mm silver layer. Protons were detected using a surface-barrier annular silicon detector at 180° . γ rays were detected using a 40 cm^3 Ge(Li) detector at 0° , 55° and 90° with FWHM=3.2 keV at $E_\gamma=1.332\text{ MeV}$ in the first experiment and a NaI crystal at 0° , 30° , 45° , 60° and 90° in the second experiment. Measured E_p , E_γ , I_γ , and $\text{p}\gamma(\theta)$ -coin. Deduced 12 levels, J, γ -ray branching ratios, and mixing ratios. Deduced $T_{1/2}$ for 7 levels using the Doppler-Shift Attenuation Method (DSAM).

1975VaYG,1972Va07: A 4.72-MeV deuteron beam of 175 nA was provided from the Groningen 5 MV Van de Graaff generator. Targets were $80\text{--}\mu\text{g}/\text{cm}^2$ Zn^{34}S enriched to 89% evaporated onto $10\text{--}\mu\text{g}/\text{cm}^2$ Formva plus $10\text{--}\mu\text{g}/\text{cm}^2$ carbon. Protons were detected using a 2-mm annular silicon detector. γ rays were detected using a $7.6\text{-cm}\times 7.6\text{-cm}$ NaI(Tl) detector. Measured E_p , E_γ , I_γ , and $\text{p}\gamma(\theta)$ -coin. Deduced levels, J, γ -ray branching ratios, and mixing ratios. Also see **1971VaZG**.

1970Bu18: 2-3.25 MeV deuteron beams were produced from the University of Arizona 5.5 MV Van de Graaff accelerator. Targets were $480\text{ }120\text{-}\mu\text{g}/\text{cm}^2$ Ag_2S made by heating 67% enriched ^{34}S powder and silver foil. Protons were detected using a 1-mm annular silicon surface barrier detectors and γ rays were detected using a $7.62\text{ cm}\times 7.62\text{ cm}$ NaI(Tl) detector in the angular correlation experiment and a Ge(Li) detector in the lifetime experiment. Measured E_p , E_γ , I_γ , and $\text{p}\gamma(\theta)$ -coin. Deduced 3 levels, J, γ -ray branching ratios, and mixing ratios. Deduced $T_{1/2}$ for 3 levels using the Doppler-shift attenuation method (DSAM).

2024Co04: A 16-MeV deuteron beam from the John D. Fox Superconducting Linear Accelerator, FSU. Particles were momentum-analyzed by the Super-Enge Split-Pole Spectrograph and identified using a position-sensitive ionization chamber with two anode wires in the isobutane gas (ΔE) and a large plastic scintillator (E) at the focal plane. γ rays were detected using a CeBr_3 Array of five CeBr_3 detectors. Measured $\text{p}\gamma$ delayed coincidence time spectra and deduced $T_{1/2}$ for the 1991 level of ^{35}S .

1971Pr11: A 4.48-MeV deuteron beam was produced from the insulated-core transformer tandem accelerator at the Aerospace Research Laboratories (ARL). The target was Ag^{25}S enriched to 85.61% in ^{34}S . Protons were detected using a $500\text{ }\mu\text{m}$ -thick, 50-mm^2 silicon detector and γ rays were detected using a 1-in.-thick, 1.5-in. in diameter NE111 scintillator. Measured $\text{p}\gamma$ -delayed coincidence time spectrum and deduced $T_{1/2}$ for the 1.99-MeV level.

 ^{35}S Levels

$E(\text{level})^\dagger$	$J^\pi^\ddagger$	$T_{1/2}$	Comments
0	$3/2^+$		
1572.5 5	$1/2^+$	$>1.3\text{ ps}$	E(level): weighted average of 1574.2 10 (1970Bu18) and 1572.3 3(1972Fr11). $T_{1/2}$: $>1.3\text{ ps}$ (1970Bu18 , $\tau>1.8\text{ ps}$). $>1.1\text{ ps}$ (1972Fr11 , $\tau>1.6\text{ ps}$).
1992.9 17	$7/2^-$	$1.03\text{ ns } 5$	E(level): unweighted average of 1994.6 10 (1970Bu18) and 1991.2 3 (1972Fr11). $T_{1/2}$: Lifetime $\tau = 1.48\text{ ns } 7$ from weighted average of $\tau = 1.7\text{ ns } 3$ (2024Co04) and $\tau = 1.47\text{ ns } 7$ (1971Pr11). Other: $\tau > 4.5\text{ ps}$ using DSAM (1970Bu18).
2348.8 13	$3/2^-$	$0.71\text{ ps } 15$	E(level): unweighted average of 2350.0 10 (1970Bu18) and 2347.5 2 (1972Fr11). $T_{1/2}$: Lifetime $\tau = 1.03\text{ ns } 21$ from weighted average of $\tau = 0.8\text{ ps } 2$ (1970Bu18) and $\tau = 1.22\text{ ps } 18$ (1972Fr11).
2718.0 10	$(3/2,5/2,7/2)$	$69\text{ fs } 24$	$T_{1/2}$: from 1972Fr11 ($\tau=100\text{ fs } 35$).
2935 3	$(3/2,5/2)$		J^π : From 1972Va07 .
3421.0 10	$(3/2,5/2,7/2)$	$<69\text{ fs}$	$T_{1/2}$: from 1972Fr11 ($\tau<100\text{ fs}$).
3558			E(level): rounded to the nearest integer from the Adopted Levels. 1972Fr11 quotes 3563 8 from 1969Mo12 without reporting their own E(level).
3592 5			
3802.0 10	$3/2^-$	$25\text{ fs } 18$	$T_{1/2}$: from 1972Fr11 ($\tau=36\text{ fs } 26$).
3815 3			
3885.0 10			
4025.0 10			
4107 2		$<55\text{ fs}$	$T_{1/2}$: from 1972Fr11 ($\tau<80\text{ fs}$).
4189 2		$<35\text{ fs}$	$T_{1/2}$: from 1972Fr11 ($\tau<50\text{ fs}$).
4480 2		$<62\text{ fs}$	$T_{1/2}$: from 1972Fr11 ($\tau<90\text{ fs}$).

† From the average of **1972Fr11** and **1970Bu18** for the lowest three excited states, and from **1972Fr11** for higher-lying states.

‡ From $\gamma(\theta)$ in **1972Fr11**, unless otherwise noted.

$^{34}\text{S}(\text{d,p}\gamma)$ **1972Fr11,1975VaYG,1970Bu18 (continued)**

$\gamma(^{35}\text{S})$								
$E_i(\text{level})$	J_i^π	$E_\gamma^\dagger$	$I_\gamma^\ddagger$	E_f	J_f^π	Mult.	δ	Comments
1572.5	1/2 ⁺	1572.5	100	0	3/2 ⁺			$A_2=-0.09\ 5$, $A_4=+0.07\ 6$ (1972Fr11); $A_2=-0.007\ 10$, $A_4=+0.004\ 20$ (1975VaYG).
1992.9	7/2 ⁻	1992.8	100	0	3/2 ⁺	(M2+E3)	+0.17 6	Mult.: D+Q or M2+E3 from $\gamma(\theta)$ and RUL; M2+E3 from level scheme. δ : unweighted average of 0.11 4 (1970Bu18), +0.11 4 (1972Fr11), and +0.28 3 (1972Va07,1975VaYG). Others: $\delta(\text{Q/D})=-1.9\ 7$ (1970Bu18) and -0.43 3 (1972Va07,1975VaYG) for $J=5/2$. $A_2=+0.35\ 4$, $A_4=-0.39\ 4$ (1972Fr11); $A_2=+0.16\ 3$, $A_4=-0.10\ 4$ (1975VaYG).
2348.8	3/2 ⁻	776.3	27 2	1572.5	1/2 ⁺	E1+M2	-0.01 4	Mult., δ : From $\gamma(\theta)$ in 1972Fr11 . δ : Other: $+0<\delta<+1.7$ (1972Va07). $A_2=-0.37\ 3$, $A_4=-0.03\ 3$ (1972Fr11); $A_2=-0.38\ 2$, $A_4=+0.01\ 4$ (1975VaYG).
		2348.7	73 2	0	3/2 ⁺	E1+M2	-0.27 12	Mult., δ : From $\gamma(\theta)$ in 1972Va07 . $A_2=+0.33\ 2$, $A_4=-0.05\ 2$ (1972Fr11); $A_2=+0.30\ 2$, $A_4=-0.02\ 4$ (1975VaYG). $B(\text{E1})(\text{W.u.})=4\times 10^{-5}$ (1972Fr11); $B(\text{E1})(\text{W.u.})=2.4\times 10^{-5}$, $B(\text{M2})(\text{W.u.})=3.9$ (1972Va07).
2718.0	(3/2,5/2,7/2)	369.2	<5	2348.8	3/2 ⁻			$A_2=+0.47\ 6$, $A_4=-0.08\ 7$ (1972Fr11); $A_2=+0.22\ 2$, $A_4=-0.02\ 3$ (1975VaYG). δ : $\delta(\text{M2/E1})>+0.5$, $B(\text{M2})(\text{W.u.})>45$ or $\delta(\text{E3/M2})=+0.18\ 18$, $B(\text{M2})(\text{W.u.})=300\ 120$ (1972Fr11).
		725.1	<5	1992.9	7/2 ⁻			
		1145.5	<5	1572.5	1/2 ⁺			
		2717.9	100	0	3/2 ⁺			
2935	(3/2,5/2)	586	<10	2348.8	3/2 ⁻			$A_2=-0.06\ 3$, $A_4=-0.01\ 6$ (1975VaYG).
		942	<20	1992.9	7/2 ⁻			
		1363	<15	1572.5	1/2 ⁺			
		2935	100	0	3/2 ⁺			
3421.0	(3/2,5/2,7/2)	486	<4	2935	(3/2,5/2)			$A_2=-0.54\ 8$, $A_4=-0.11\ 7$ (1972Fr11).
		703.0	<6	2718.0	(3/2,5/2,7/2)			
		1072.2	<7	2348.8	3/2 ⁻			
		1428.1	<30	1992.9	7/2 ⁻			
		1848.5	<7	1572.5	1/2 ⁺			$A_2=-0.54\ 8$, $A_4=-0.11\ 7$ (1972Fr11).
		3420.8	100	0	3/2 ⁺			
3592		657	<10	2935	(3/2,5/2)			
		874	<11	2718.0	(3/2,5/2,7/2)			
		1243	<15	2348.8	3/2 ⁻			$A_2=-0.54\ 8$, $A_4=-0.11\ 7$ (1972Fr11).
		1599	<17	1992.9	7/2 ⁻			
		2019	<17	1572.5	1/2 ⁺			
		3592	100	0	3/2 ⁺			
3802.0	3/2 ⁻	867	<4	2935	(3/2,5/2)			

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$^{34}\text{S}(\text{d},\text{p}\gamma)$ **1972Fr11,1975VaYG,1970Bu18** (continued) $\gamma(^{35}\text{S})$ (continued)

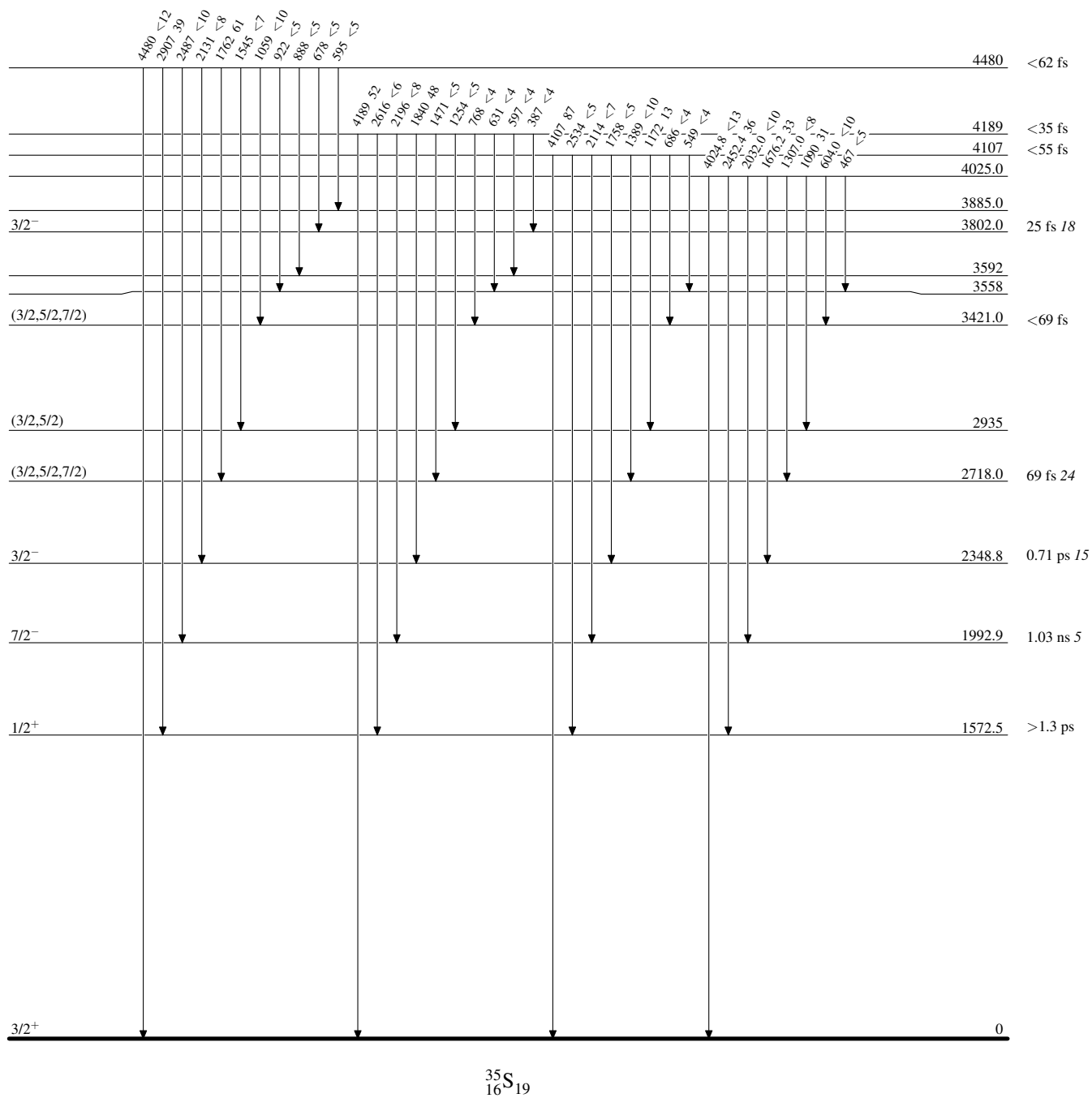
$E_i(\text{level})$	$E_\gamma^\dagger$	$I_\gamma^\ddagger$	E_f	J_f^π	Comments
3802.0	1084.0	5 2	2718.0	(3/2,5/2,7/2)	
	1453.2	8 3	2348.8	3/2 ⁻	
	1809.1	<4	1992.9	7/2 ⁻	
	2229.4	42 4	1572.5	1/2 ⁺	$A_2=-0.56$ I5, $A_4=0.0$ I (1972Fr11).
	3801.8	45 4	0	3/2 ⁺	$A_2=+0.19$ I2, $A_4=0.0$ I (1972Fr11).
3815	1822	100	1992.9	7/2 ⁻	
3885.0	464.0	<30	3421.0	(3/2,5/2,7/2)	
	950	<12	2935	(3/2,5/2)	
	1167.0	<15	2718.0	(3/2,5/2,7/2)	
	1536.2	<18	2348.8	3/2 ⁻	
	1892.1	100	1992.9	7/2 ⁻	
	2312.4	<16	1572.5	1/2 ⁺	
	3884.8	<16	0	3/2 ⁺	
4025.0	467	<5	3558		
	604.0	<10	3421.0	(3/2,5/2,7/2)	
	1090	31 8	2935	(3/2,5/2)	
	1307.0	<8	2718.0	(3/2,5/2,7/2)	
	1676.2	33 12	2348.8	3/2 ⁻	
	2032.0	<10	1992.9	7/2 ⁻	
	2452.4	36 12	1572.5	1/2 ⁺	
	4024.8	<13	0	3/2 ⁺	
4107	549	<4	3558		
	686	<4	3421.0	(3/2,5/2,7/2)	
	1172	13 6	2935	(3/2,5/2)	
	1389	<10	2718.0	(3/2,5/2,7/2)	
	1758	<5	2348.8	3/2 ⁻	
	2114	<7	1992.9	7/2 ⁻	
	2534	<5	1572.5	1/2 ⁺	
	4107	87 6	0	3/2 ⁺	
	4189	<4	3802.0	3/2 ⁻	
4189	597	<4	3592		
	631	<4	3558		
	768	<4	3421.0	(3/2,5/2,7/2)	
	1254	<5	2935	(3/2,5/2)	
	1471	<5	2718.0	(3/2,5/2,7/2)	
	1840	48 5	2348.8	3/2 ⁻	
	2196	<8	1992.9	7/2 ⁻	
	2616	<6	1572.5	1/2 ⁺	
	4189	52 5	0	3/2 ⁺	
	4480	<5	3885.0		
	678	<5	3802.0	3/2 ⁻	
4480	888	<5	3592		
	922	<5	3558		
	1059	<10	3421.0	(3/2,5/2,7/2)	
	1545	<7	2935	(3/2,5/2)	
	1762	61 7	2718.0	(3/2,5/2,7/2)	
	2131	<8	2348.8	3/2 ⁻	
	2487	<10	1992.9	7/2 ⁻	
	2907	39 7	1572.5	1/2 ⁺	
	4480	<12	0	3/2 ⁺	

[†] Deduced by evaluators from level-energy difference.[‡] From 1972Fr11, unless otherwise noted.

$^{34}\text{S}(\text{d},\text{p}\gamma)$ 1972Fr11,1975VaYG,1970Bu18

Level Scheme

Intensities: % photon branching from each level



³⁴S(d,p γ)

1972Fr11,1975VaYG,1970Bu18

Level Scheme (continued)

Intensities: % photon branching from each level

